

Chriss (1999, 2000) that finds that trading strategy (style) influences trading cost. Breen, Hodrick and Korajczyk (2002) provide a foundation for testing models which we utilize in the model error analysis section of the scientific method (Step 5: Analyze the Data).

From these publications a common underlying theme: price impact is caused by order size, trading strategy (e.g., level of transaction urgency, percentage of volume, participation rate, etc.), volatility, market capitalization, side (buy/sell)—as well as spreads and price.

Next we observe and analyze actual customer order data. Our data observation universe consists of actual executed trades during a three month period Jan. 2010 through Mar. 2010. While there are no specific steps to observe and analyze data, we recommend visualization by plotting data and simple analyses such as linear regression to uncover potential relationships.

As part of this step, we plotted the average trading cost (measured as the difference between the average execution price and mid-point of the spread at the time of the order arrival) as a function of several different variables including size, volatility, percentage of volume (POV) rate, and price. We segmented data into large cap and small cap stock (as per our literature research findings). Stocks with market capitalization of \$2B or more were classified as large cap stocks, while stocks with market capitalization less than \$2B were classified as small cap stocks. Traditionally, large cap stocks are categorized as stocks with a market cap greater than \$5B, mid-cap stocks as stocks with a market cap between \$2B and \$5B, and small cap stocks as stocks with market caps less than \$2B. We grouped mid-cap stocks in the large cap or small cap categories based on actual market cap. Readers are welcome to repeat steps in this chapter to determine if an additional category of stocks is needed.

There are several issues worth mentioning:

- First, share amounts traded were not necessarily the entire order. We were not able to observe the number of unexecuted shares or opportunity cost corresponding to the total order size. This may lead to survivorship bias where orders with favorable price momentum are completed more often than orders with adverse market movement resulting in lower than actual observed costs.
- Second, we were unable to observe actual specified trading strategy at the beginning of trading. We do not know whether traders or managers engaged in any opportunistic trading during the execution of the order. This would occur in situations whereby traders became more